

API Security Risk Analysis Report

Future Interns – Cyber Security Internship

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Introduction:

This report presents an API Security Risk Analysis conducted on the JSON Placeholder public API. The objective was to review API endpoints, inspect response headers, evaluate authentication requirements, and identify potential security risks. The assessment was performed using Insomnia in a read-only and ethical manner.

API Information:

API Name: JSONPlaceholder

API URL: <https://jsonplaceholder.typicode.com/posts>

Testing Tool: Insomnia

Request Method: GET

Methodology:

The API endpoint was tested using Insomnia. Response data and headers were inspected to evaluate authentication mechanisms, exposed data, and other observable security controls. No exploitation or unauthorized testing was performed.

The screenshot displays the Insomnia 13.0.3 application window. The top bar shows the application name and version, along with standard menu items (Application, File, Edit, View, Window, Tools, Help) and a search bar. The left sidebar contains a 'Projects' section with a 'Scratch Pad' project, and a 'Collections' section with a 'Documents' collection. The main workspace is divided into two panes. The left pane shows a REST client request configuration for a GET request to 'https://jsonplaceholder.typicode.com/posts'. The right pane shows the response, which is a JSON array of five post objects. The response status is '200 OK' and the response time is '454 ms'. The response body is a JSON array of five objects, each representing a post with fields like 'id', 'userId', 'title', and 'body'.

Insomnia 13.0.3
Application File Edit View Window Tools Help

Star 39,082 Search... Ctrl + P Login Sign up for free

Projects

- Scratch Pad > New Request

GET https://jsonplaceholder.typicode.com/posts Send 200 OK 454 ms 26.9 KB Just Now

Params Body Auth Headers 3 Scripts Docs

URL PREVIEW
https://jsonplaceholder.typicode.com/posts

QUERY PARAMETERS
+ Add - Delete all - Description Import from URL Bulk Edit

id	name	value
1	name	value

PATH PARAMETERS
Path parameters are url path segments that start with a colon ':' e.g. 'id'

Preview Headers 22 Cookies Tests 0/0 Mock Console

```
1 [
2   {
3     "userId": 1,
4     "id": 1,
5     "title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",
6     "body": "quia et suscipit\nvixit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas tota\nnostrum rerum est autem sunt rem eveniet architecto"
7   },
8   {
9     "userId": 1,
10    "id": 2,
11    "title": "qui est esse",
12    "body": "et rerum temporum\nvitae\nsequi sint nihil reprehenderit dolor beatae ea dolores neque\nfugiat blanditiis voluptate porro vel nihil molestiae ut reiciendis\nqui aperiam non debitis possimus qui neque nisi nulla"
13  },
14  {
15    "userId": 1,
16    "id": 3,
17    "title": "ea molestias quasi exercitationem repellat qui ipsa sit aut",
18    "body": "et justo sed quo\nirene\nvoluptatem occaecati omnis eligendi aut ad\nvoluptatem doloribus vel accusantium quis pariatur\nmolestiae porro eius odio et labore et velit aut"
19  },
20  {
21    "userId": 1,
22    "id": 4,
23    "title": "eum et est occaecati",
24    "body": "laudantium ut quasi asperiores repellat est ut et veniam fugiat sit ut"
25  },
26  {
27    "userId": 1,
28    "id": 5,
29    "title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",
30    "body": "temporibus autem et autem fugiat quos sint consequuntur asperiores sunt optio delectus harum quia voluptatem sit reprehenderit culpa"
31  }
32 ]
```

\$.store.books[0].author

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Mocks
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36°C Partly sunny
ENG IN 02:47 PM 20-06-2026

The screenshot displays the Insomnia 13.0.1 application window. The top bar shows the application name and version, along with standard menu items (Application, File, Edit, View, Window, Tools, Help) and a search bar. The left sidebar contains a 'Projects' section with a filter, a list of collections (Insomnia, Scratch Pad), and a 'New Request' button. The main workspace is divided into three panes. The top pane shows a REST client request to 'https://jsonplaceholder.typicode.com/posts' with a 'Send' button. The middle pane displays the 'Headers' tab, showing a table of request headers. The bottom pane shows the 'Body' tab, which is currently empty. The right sidebar contains a 'Preview' tab, showing a table of response headers. The bottom status bar displays system information (36°C, Partly sunny) and the time (02:50 PM, 20-06-2026).

Application: Insomnia 13.0.1

Menu: Application File Edit View Window Tools Help

Search: Search...

Buttons: Login Sign up for free

Projects: Filter

Collection: Insomnia

Item: Scratch Pad

Request: GET New Request

Request Details: GET https://jsonplaceholder.typicode.com/posts Send 200 OK 454 ms 26.9 KB 4 Minutes Ago

Params: Body Auth Headers (3) Scripts Docs

URL PREVIEW: https://jsonplaceholder.typicode.com/posts

QUERY PARAMETERS: Add Delete all Description Import from URL Bulk Edit

Headers: 22 Cookies Tests 0/0 Mock Console

Header Table:

NAME	VALUE
date	Sat, 20 Jun 2026 09:16:53 GMT
content-type	application/json; charset=utf-8
access-control-allow-credentials	true
cache-control	max-age=43200
etag	W/"6b80-Ybsq/K6GwwqYkAsFqDXGC7DoM"
expires	-1
nel	{"report_to":"heroku-nel","response_headers":{"Via"},"max_age":3600,"success_fraction":0.01,"failure_fraction":0.1}
pragma	no-cache
report-to	{"group":"heroku-nel","endpoints":[{"url":"https://nel.heroku.com/reports?ts=178184%2Bsd5xq8iBwGiq4%2Bia9%2FYA7W2AZC0MOIFMVZk%3D&sid=e11707d5-02a7-43ef-b45e-2c4d2036f7d0&id=1781931529"}],"max_age":3600}
reporting-endpoints	heroku-nel="https://nel.heroku.com/reports?ts=178184%2Bsd5xq8iBwGiq4%2Bia9%2FYA7W2AZC0MOIFMVZk%3D&sid=e11707d5-02a7-43ef-b45e-2c4d2036f7d0&id=1781931529"
server	cloudflare

PATH PARAMETERS: Path parameters are url path segments that start with a colon ':' e.g. ':id'

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System: 36°C Partly sunny 02:50 PM 20-06-2026

Security Risks Identified:

Risk 1: No Authentication Required

Severity: Medium

The API endpoint can be accessed without requiring authentication, API keys, or user credentials. While this behavior is acceptable for a public testing API, similar configurations in production environments could expose sensitive data to unauthorized users.

Risk 2: Open Data Exposure

Severity: Low

The API returns data directly to any user accessing the endpoint. Although the information provided is test data, unrestricted access to real business data could lead to information disclosure risks.

Risk 3: No Visible Rate Limiting

Severity: Medium

No visible rate-limiting mechanism was observed during testing. Without rate limiting, attackers may generate excessive requests that could impact application performance and service availability.

Business Impact:

Weak API security controls can increase the risk of unauthorized access, information disclosure, service abuse, and reputational damage.

Organizations should regularly review API configurations and security controls to protect sensitive information and maintain service reliability.

Remediation Recommendations:

- Implement authentication for sensitive API endpoints.
- Apply role-based access controls.
- Limit exposure of unnecessary data.
- Implement API rate limiting.
- Monitor API access logs regularly.

- Conduct periodic API security assessments.

Conclusion:

This API Security Risk Analysis identified common API security considerations, including unauthenticated access, open data exposure, and the absence of visible rate-limiting controls. The assessment demonstrates how security analysts evaluate APIs, identify risks, and recommend security improvements in a professional and ethical manner.